

Troca de Chave	Algoritmo	Latência do Handshake [ms]					
		Desligado	Padrão(3x)	5x	10x	15x	20x
BP384 + ML-KEM-768	P384 + ML-DSA-65	10.1±0.4	16.4±0.9	-	-	-	-
BP384 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	20.6±0.3	32.1±8.0	-	-	-	-
BP384 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	221.5±1.1	264.7±5.2	-	-	-	-
BP384 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	31.6±0.4	63.0±1.2	-	-	-	-
BP384 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	463.4±1.0	463.1±1.3	-	-	-	-
P256 + ML-KEM-768	P384 + ML-DSA-65	4.7±0.5	6.3±1.4	-	-	-	-
P256 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	15.1±0.5	29.7±0.9	-	-	-	-
P256 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	215.7±1.3	243.2±17.5	-	-	-	-
P256 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	26.0±0.6	50.6±7.1	-	-	-	-
P256 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	456.6±0.9	556.1±50.2	-	-	-	-
P384 + ML-KEM-768	P384 + ML-DSA-65	9.8±0.3	14.9±1.5	-	-	-	-
P384 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	20.3±0.1	32.5±6.7	-	-	-	-
P384 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	220.3±0.3	279.9±50.7	-	-	-	-
P384 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	31.0±0.1	57.0±9.0	-	-	-	-
P384 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	460.5±0.2	658.2±206.9	-	-	-	-
P384 + BIKE-L3	P384 + ML-DSA-65	12.0±0.4	16.7±2.4	-	-	-	-
P384 + BIKE-L3	P384 + SPHINCS-SHA2-192-F-SIMPLE	22.4±0.5	35.3±4.9	-	-	-	-
P384 + BIKE-L3	P384 + SPHINCS-SHA2-192-S-SIMPLE	222.1±0.2	249.6±12.8	-	-	-	-
P384 + BIKE-L3	P384 + SPHINCS-SHAKE-192-F-SIMPLE	33.7±0.8	56.9±6.1	-	-	-	-
P384 + BIKE-L3	P384 + SPHINCS-SHAKE-192-S-SIMPLE	463.2±0.7	488.3±14.6	-	-	-	-
P384 + FRODO-976-AES	P384 + ML-DSA-65	14.9±2.4	1020.6±2.7	-	-	-	-
P384 + FRODO-976-AES	P384 + SPHINCS-SHA2-192-F-SIMPLE	31.5±6.8	1046.3±5.2	-	-	-	-
P384 + FRODO-976-AES	P384 + SPHINCS-SHA2-192-S-SIMPLE	222.8±1.3	1268.5±5.2	-	-	-	-
P384 + FRODO-976-AES	P384 + SPHINCS-SHAKE-192-F-SIMPLE	34.5±1.3	1065.6±4.9	-	-	-	-
P384 + FRODO-976-AES	P384 + SPHINCS-SHAKE-192-S-SIMPLE	464.0±2.1	1508.6±7.8	-	-	-	-
P384 + FRODO-976-SHAKE	P384 + ML-DSA-65	20.2±3.5	1027.4±3.4	-	-	-	-
P384 + FRODO-976-SHAKE	P384 + SPHINCS-SHA2-192-F-SIMPLE	28.7±5.5	1047.0±7.2	-	-	-	-
P384 + FRODO-976-SHAKE	P384 + SPHINCS-SHA2-192-S-SIMPLE	227.3±2.4	1270.6±2.4	-	-	-	-
P384 + FRODO-976-SHAKE	P384 + SPHINCS-SHAKE-192-F-SIMPLE	37.0±1.4	1072.9±4.5	-	-	-	-
P384 + FRODO-976-SHAKE	P384 + SPHINCS-SHAKE-192-S-SIMPLE	467.0±1.9	1512.2±6.5	-	-	-	-
P384 + HQC-192	P384 + ML-DSA-65	81.6±3.9	69386.8±24010.2	-	-	-	-
P384 + HQC-192	P384 + SPHINCS-SHA2-192-F-SIMPLE	91.2±0.6	-	-	-	-	-
P384 + HQC-192	P384 + SPHINCS-SHA2-192-S-SIMPLE	295.1±2.8	88296.7±2684.4	-	-	-	-
P384 + HQC-192	P384 + SPHINCS-SHAKE-192-F-SIMPLE	117.0±16.5	80718.1±16650.7	-	-	-	-
P384 + HQC-192	P384 + SPHINCS-SHAKE-192-S-SIMPLE	541.9±4.5	87056.5±2826.0	-	-	-	-
X25519 + ML-KEM-768	P384 + ML-DSA-65	5.0±0.5	7.1±1.4	-	-	-	-
X25519 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	15.4±0.7	48.5±19.5	-	-	-	-
X25519 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	215.1±0.6	252.6±36.6	-	-	-	-
X25519 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	25.6±0.3	48.7±6.6	-	-	-	-
X25519 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	455.7±0.3	524.1±35.9	-	-	-	-
X448 + ML-KEM-768	P384 + ML-DSA-65	6.0±0.7	7.5±1.9	-	-	-	-
X448 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	15.7±0.3	28.4±4.0	-	-	-	-
X448 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	216.2±0.7	261.7±28.7	-	-	-	-
X448 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	26.7±0.8	43.5±12.2	-	-	-	-
X448 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	456.6±0.3	605.5±141.8	-	-	-	-
X448 + BIKE-L3	P384 + ML-DSA-65	7.2±0.2	7.7±0.7	-	-	-	-
X448 + BIKE-L3	P384 + SPHINCS-SHA2-192-F-SIMPLE	18.1±0.5	30.9±4.8	-	-	-	-
X448 + BIKE-L3	P384 + SPHINCS-SHA2-192-S-SIMPLE	217.8±0.6	256.7±23.4	-	-	-	-
X448 + BIKE-L3	P384 + SPHINCS-SHAKE-192-F-SIMPLE	29.9±0.7	46.9±11.0	-	-	-	-
X448 + BIKE-L3	P384 + SPHINCS-SHAKE-192-S-SIMPLE	459.5±1.1	496.0±10.9	-	-	-	-
X448 + FRODO-976-AES	P384 + ML-DSA-65	7.8±0.6	1010.0±0.3	-	-	-	-
X448 + FRODO-976-AES	P384 + SPHINCS-SHA2-192-F-SIMPLE	27.3±0.8	1029.9±4.2	-	-	-	-
X448 + FRODO-976-AES	P384 + SPHINCS-SHA2-192-S-SIMPLE	218.8±2.0	1258.0±23.9	-	-	-	-
X448 + FRODO-976-AES	P384 + SPHINCS-SHAKE-192-F-SIMPLE	28.2±0.4	1053.6±2.4	-	-	-	-
X448 + FRODO-976-AES	P384 + SPHINCS-SHAKE-192-S-SIMPLE	459.0±1.4	1499.3±8.8	-	-	-	-
X448 + FRODO-976-SHAKE	P384 + ML-DSA-65	13.2±0.5	1017.2±1.2	-	-	-	-
X448 + FRODO-976-SHAKE	P384 + SPHINCS-SHA2-192-F-SIMPLE	28.6±6.4	1041.9±6.2	-	-	-	-
X448 + FRODO-976-SHAKE	P384 + SPHINCS-SHA2-192-S-SIMPLE	221.2±1.3	1264.1±5.4	-	-	-	-
X448 + FRODO-976-SHAKE	P384 + SPHINCS-SHAKE-192-F-SIMPLE	34.1±1.0	1062.8±6.1	-	-	-	-
X448 + FRODO-976-SHAKE	P384 + SPHINCS-SHAKE-192-S-SIMPLE	463.5±1.4	1507.8±6.9	-	-	-	-
X448 + HQC-192	P384 + ML-DSA-65	74.2±0.4	75181.7±8153.9	-	-	-	-
X448 + HQC-192	P384 + SPHINCS-SHA2-192-F-SIMPLE	86.7±1.6	80683.5±5649.8	-	-	-	-
X448 + HQC-192	P384 + SPHINCS-SHA2-192-S-SIMPLE	293.0±10.2	80505.9±3807.4	-	-	-	-
X448 + HQC-192	P384 + SPHINCS-SHAKE-192-F-SIMPLE	97.4±1.7	81280.4±1624.2	-	-	-	-